

25 The ore iron(II) chromite, FeCr_2O_4 , is mined in the Philippines and Afghanistan.

The first stage in the production of chromium from chromite is reaction with oxygen in molten sodium hydroxide to form sodium chromate, Na_2CrO_4 . You should ignore the oxidation of iron in this reaction.

How many moles of oxygen molecules are required to react completely with one mole of chromite ions?

A 1

B 1.5

C 3

D 6